

SRv6 Automation with Crosswork Network Controller

From path programming to an operating model for intent, visibility,
and closed-loop control



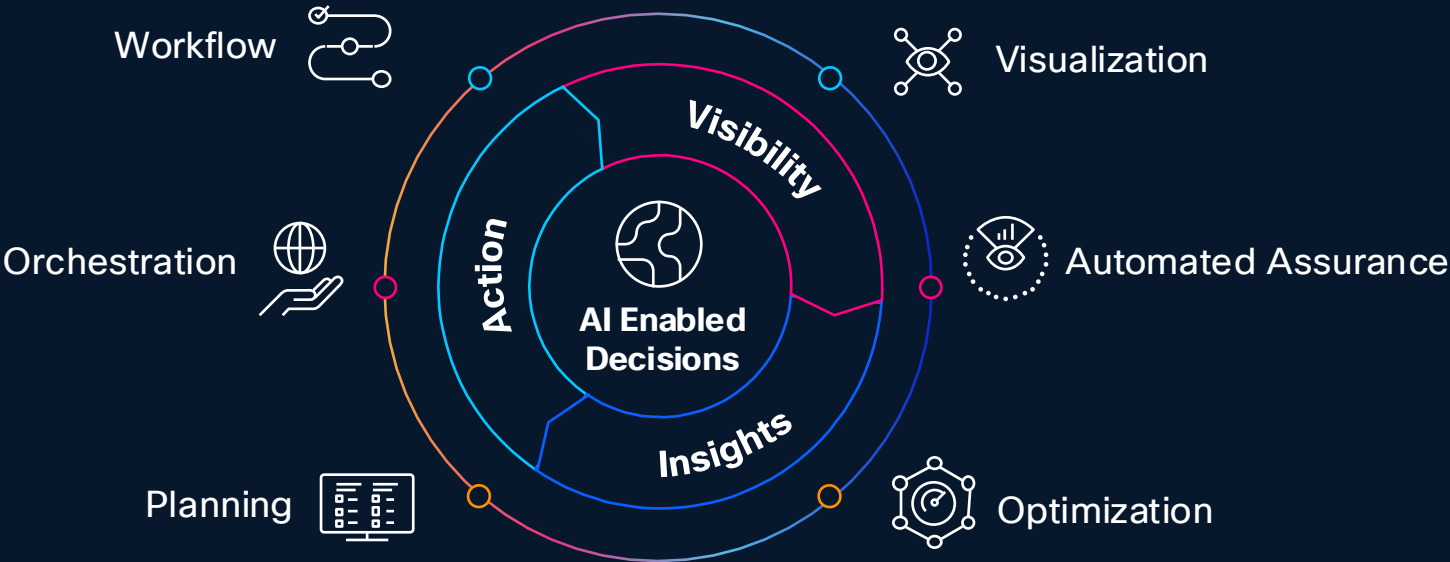
Agenda

- 01 SRv6 Automation in CNC today
- 02 SRv6 Bandwidth Management Strategy
- 03 SRv6 Observability Tools
- 04 SRv6 Congestion Mitigation
- 05 Key Takeaways and Next Steps

Crosswork Overview: Automation and Assurance

AI-enabled, multi-vendor automation of the complete lifecycle of network operations

-  Most Comprehensive Functions
-  Intent-driven Automation
-  Modular, Multi-vendor and Extensible
-  Multi-agentic AI
-  Low-code/ No-code Experience
-  Scalable, Resilient and Secure
-  Deep Visibility and Analytics



Why SRv6 Automation Matters

How to move from a programmable data plane to a repeatable service lifecycle

01

Speed

Reduce manual touch points when services and policies change

02

Scale

Apply the same operating model across domains, services, and devices

03

Reliability

Validate what was intended and see what the network is doing

04

Agility

Use traffic and performance feedback to optimize the next action

SRv6 lets the network express path intent in the packet. Operators still need a system that can turn intent into validated, observable, and adaptable behavior.

SRv6 Automation: Use Cases for Crosswork Network Controller

Service provisioning

Activate L2 / L3 services with SRv6 steering through a repeatable workflow

Traffic engineering

Compute paths with explicit constraints, Flex-Algo, or path diversity

Observability

Use Network Visibility, Demand Matrix and Capacity Planning to understand Traffic, Demands and their potential evolutions

Congestion mitigation

Leverage demand awareness to compute the most efficient detour paths

Troubleshooting

Correlate path and performance signals for improved visibility

Migration

Evolve from SR-MPLS or mixed transport while preserving service continuity

SRv6 Real-time Network Visualization

SRv6 Policies

ECMP Paths

Crosswork Network Controller

SRv6 Policies

Give feedback

Traffic Engineering

Show: Traffic Engineering

Device groups: Location

Last update: 18-Sep-2026 12:16:38 PM EDT

Refresh

Saved views

Select a saved view

Save view

Topology

Services & Traffic Engineering

Workflow Automation

Device Management

Alerts

Administration

Location

Show: ☐ Participating only ☐ IGP path

Map of the United States showing network nodes and connections.

Traffic engineering

Refined by: Headend or endpoint

SR-MPLS SRv6 Tree-SID RSVP-TE

0 Admin down 12 Oper up 0 Oper down

+ Create

Selected 6 / Total 12

Head...	Endp...	Color	Admin ...	Oper st...	Actions
☒	node-4	node-3	600	☒	☒
☒	node-5	node-3	600	☒	☒
☒	node-3	node-4	600	☒	☒
☐	node-5	node-4	600	☒	☒
☐	node-4	node-5	600	☒	☒
☒	node-3	no		☒	☒
☐	node-5	no		☒	☒

Crosswork Network Controller

ECMP Paths

Give feedback

Traffic Engineering

Show: Traffic Engineering

Device groups: Location

Last update: 18-Sep-2026 12:19:09 PM EDT

Refresh

Saved views

Select a saved view

Save view

Topology

Services & Traffic Engineering

Workflow Automation

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Show: ☐ Participating only ☒ IGP path

Map of the United States showing network nodes and connections.

Traffic engineering

Refined by: Headend or endpoint

SR-MPLS SRv6 Tree-SID RSVP-TE

0 Admin down 12 Oper up 0 Oper down

+ Create

Selected 1 / Total 12

Head...	Endp...	Color	Admin ...	Oper st...	Actions
☐	node-4	node-3	600	☒	☒
☐	node-5	node-3	600	☒	☒
☐	node-3	node-4	600	☒	☒
☒	node-5	node-4	600	☒	☒
☐	node-4	node-5	600	☒	☒
☐	node-3	node-5	600	☒	☒

Policy Details

Candidate Path details

SRv6 policy details

Current History

Headend node-5 | Source IP: 2010::5
TE RID: 198.19.1.5 | IPv6 RID: 2010::5
PCC IP: 2010::5

Endpoint node-4 | Dest IP: 2010::4
TE RID: 198.19.1.4 | IPv6 RID: 2010::4

Color 600

Summary

Admin state ☒ Up

Oper state ☒ Up

Binding SID fcbb:0:5:e009:: Behavior - 99

Policy type Regular

Profile ID -

Traffic rate -

Unused True

Delay 14

See more

Candidate path

Path name Preference Path type State

☒ bgp_c_600_ep_2010... 100 Unknown ☒ A

Segment Weight 1

Segm... Segm... SID Behavior Algo Address

0 uN fcbb:0:4::/48 End with ... 0 2010::4

Path name bgp_c_600_ep_2010::4_discr_100

Oper state ☒ Up ☒ Active

Metric type IGP

Bandwidth -

Disjoint group ID:

Association source: -

Type: -

✓

Visibility is paramount

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SRv6 Bandwidth Management Strategy

SRv6 Bandwidth Optimization Strategy



Observability

- Real-time visibility into live traffic flows & congestion hotspots
- Unified context combining topology, telemetry, and event feeds
- Optimize bandwidth utilization
- Identify risks in topology, routing, and traffic engineering



Insights and Analysis

- Predictive, real-time capacity management
- Proactively assess vulnerabilities and risks
- Identify hot-spots and capacity gaps early
- Minimize the need for congestion mitigation

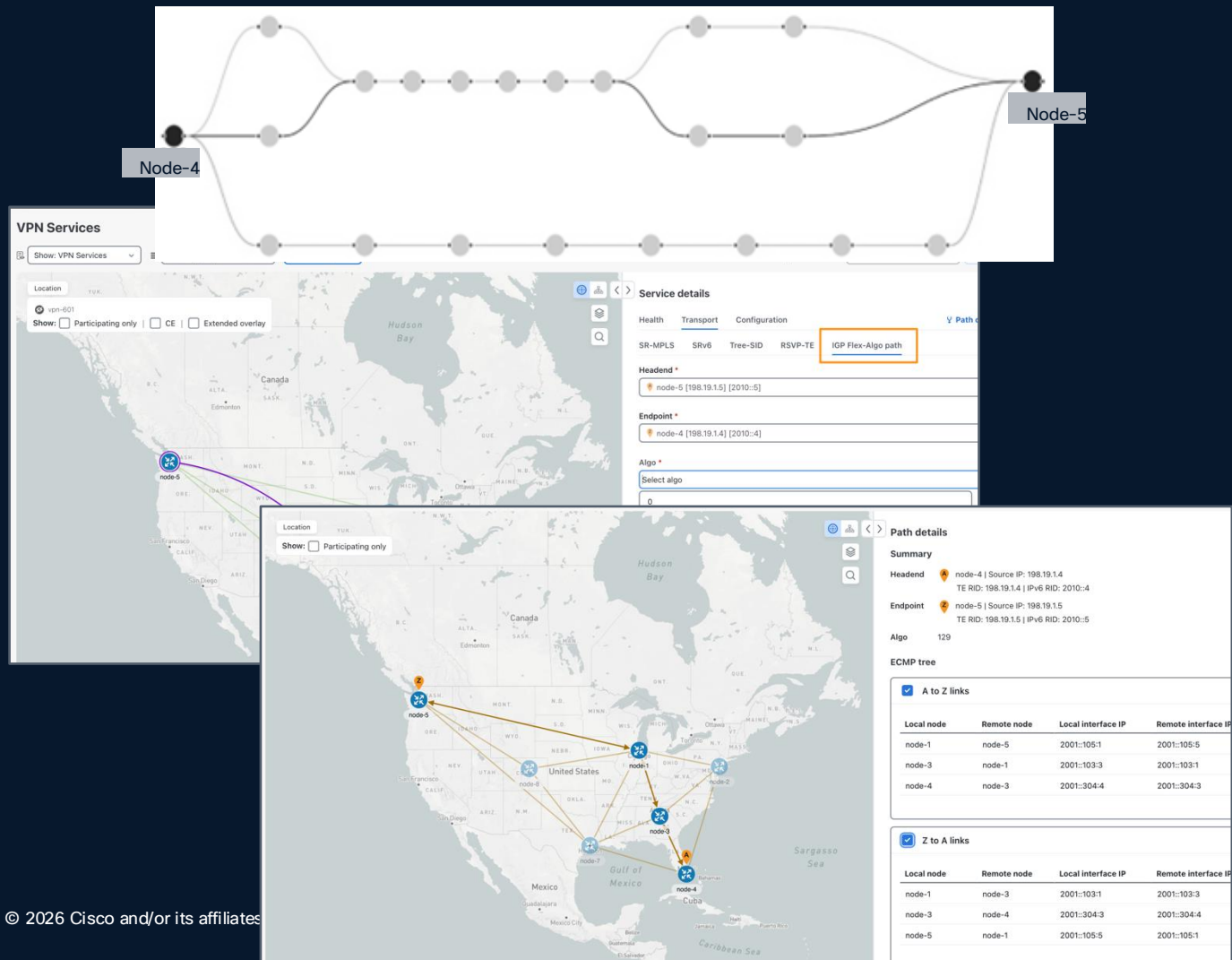


Congestion Mitigation

- Last-resort solution for unexpected events
- Mitigate failures and congestion
- Use tactical TE for congestion mitigation

Observe the Network Before Acting

Routing Analytics and Integrated Performance Measurement connect path behavior to user experience



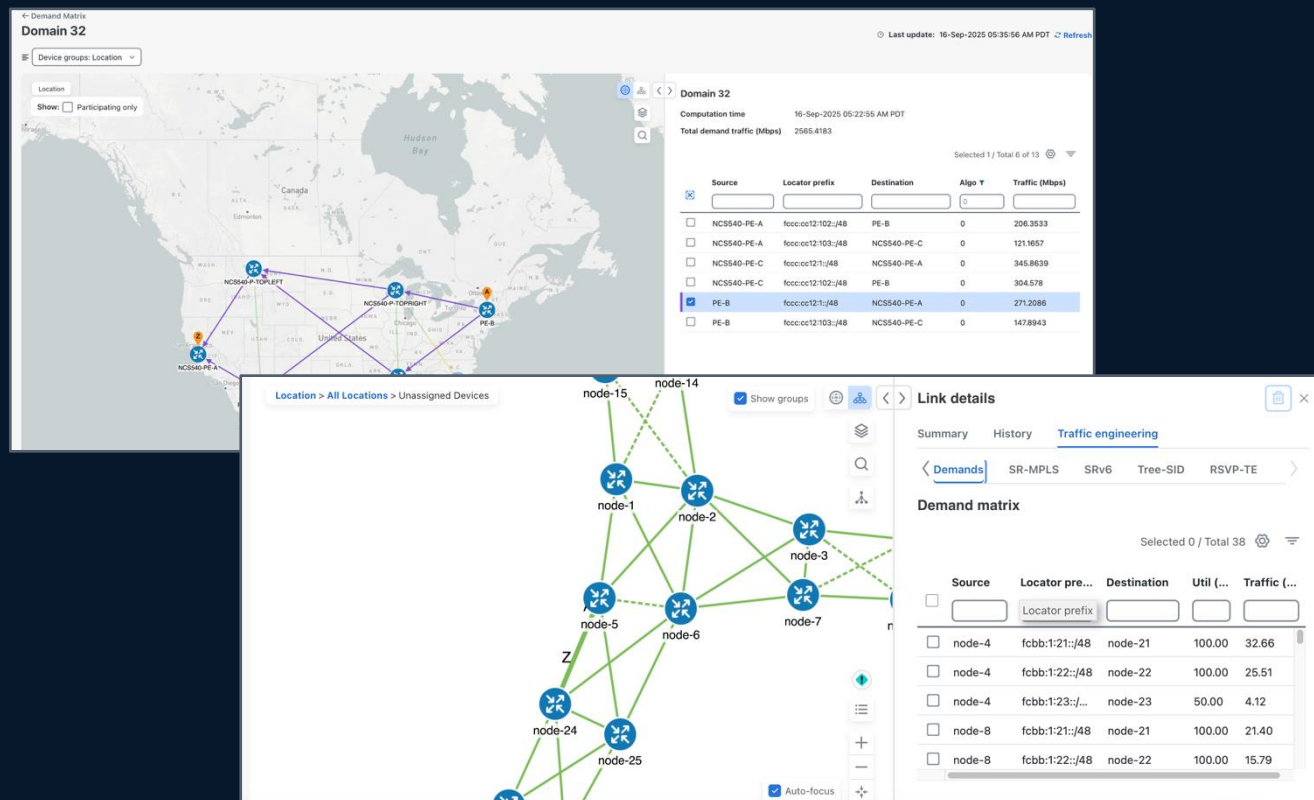
Routing Analytics

- **Path visibility**
See the ECMP path sets between source and destination
- **Change awareness**
Track path changes and stability over time
- **Performance context**
Correlate path data with loss, latency, and liveness

If you cannot see the path, you cannot explain the symptom.

Traffic Visibility and Capacity Analysis

Know where the traffic is going before capacity becomes a problem



Deterministic Demand Matrix

The demand matrix turns hardware-based SRv6 counters into an end-to-end view of traffic between nodes.

DDM

What is moving, where, and how it changes

ACP

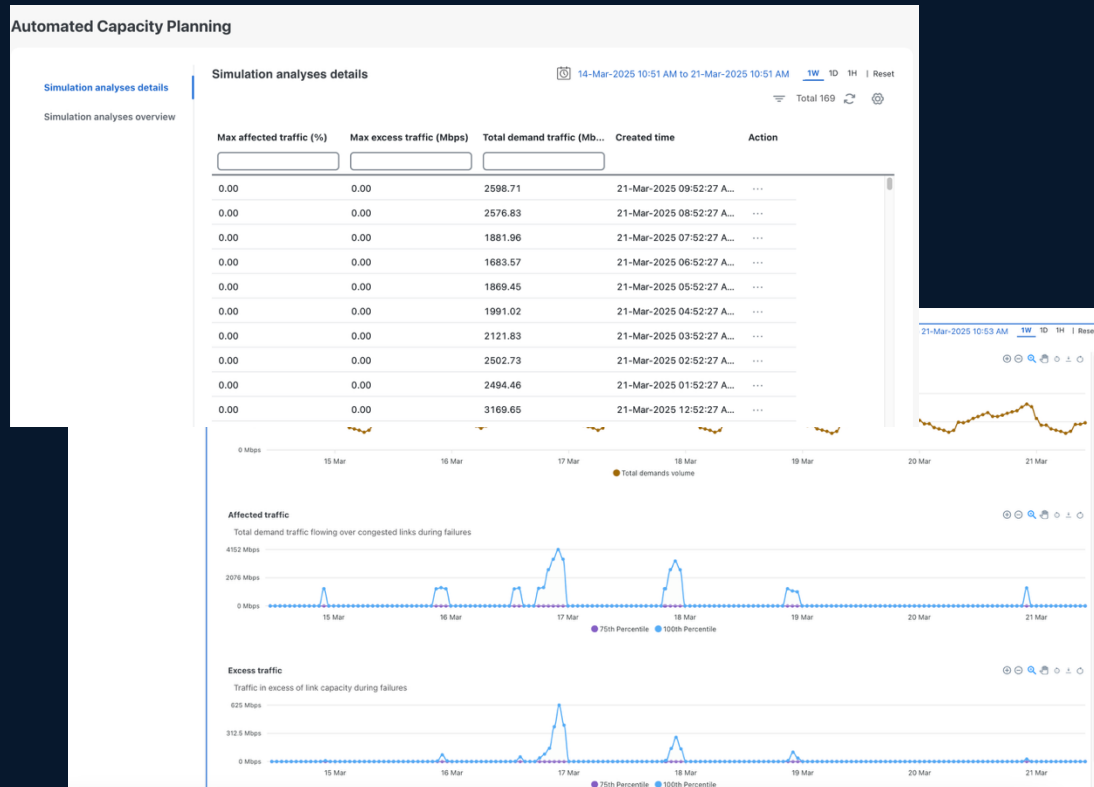
Which hotspots or failure scenarios deserve attention

Capacity planning becomes continuous input, not a periodic snapshot

The point is anticipation: identify a concern early enough to choose a policy, capacity, or maintenance response.

SRv6 Innovation: Automated Capacity Planning

Identify traffic bottlenecks or hotspots in real-time



Challenges

- Need to identify traffic bottlenecks or hotspots in real-time
- Need to anticipate future bandwidth needs

Solution

- Collects DDM counters from all edge routers (telemetry)
- Simulate what-if scenarios (link/node failure, growth)
- Network Utilization metrics: how hot the network is
- Highlights need for capacity upgrade

Outcome

- Quickly identify potential bottlenecks
- Predict potential congestion or traffic loss
- Quantify potential customer impact of link or node failure

Introducing Demand Matrix in CNC

- Track the changing traffic patterns across entire network
- Characterize the traffic flows and volumes

Crosswork Network Controller

Dashboard

Topology

Services & Traffic Engineering

Device Management

Alerts

Administration

Services & Traffic Engineering

Transport Slices

Slice Template Catalog

VPN Services

Service Health Dashboard

VPN TE Management

Traffic Engineering

TE Dashboard

Bandwidth on Demand

Local Congestion Mitigation

Circuit Style SR-TE

Demand Matrix

Path Query

Demand Matrix

Domain 32

Device groups: Location

Location

Show: ☐ Participating only

Demand Matrix

Domain 32

Device groups: Location

Location

Link details

Summary History Traffic engineering

General Demands SR-MPLS SRv6 Tree-SID RSVP-TE

Selected 0 / Total 6

	Source	Locator prefix	Destination	ECMP split (%)	Traffic (Mbps)
☐					
☐	PE-B	fcc:cc12:1::/48	NCS540-PE-A	25.00	67.80225
☐	NCS540-PE-C	fcc:cc12:1::/48	NCS540-PE-A	25.00	86.466
☐	PE-B	fcc:cc13:1::/48	NCS540-PE-A	100.00	178.529
☐	NCS540-PE-A	fcc:cc12:102::/48	PE-B	25.00	51.58825
☐	NCS540-PE-A	fcc:cc13:102::/48	PE-B	100.00	205.165
☐	NCS540-PE-A	fcc:cc12:103::/48	NCS540-PE-C	25.00	30.2915

Building the Deterministic Demand Matrix (DDM)



- Collect counters from all nodes in the domain
 - Ongoing IETF standardization for counters (draft-ietf-spring-srv6-yang, draft-ietf-spring-sr-policy-yang)
- Obtain topology from BGP-LS route reflector
- Derive the volume of each demand and SR policy

DDM as a Foundational Building Block

to \ from	A	B	C	D
A	-	5G	18G	20G
B	3G	-	21G	19G
C	15G	23G	-	7G
D	24G	17G	4G	-



Deterministic Demand Matrix



Automated Capacity Planning



Bandwidth Optimization



Coordinated Maintenance

And much more...

SRv6 is the opportunity to introduce this game-changing solution.

Leverage DDM to compute effective alternate paths when congestion occurs

-
- Diagram illustrating a network topology and a specific path from node A to node D.
- The network consists of nodes A, B, C, D, E, F, G, and H, connected by links. A central component labeled "Core" is shown at the bottom.
- A path from A to D is highlighted in red and blue. The path starts at A, goes through E, G, and D. The segment from A to E is red, and the segment from E to G is blue. The segment from G to D is also blue. The segment from A to B is dashed blue, and the segment from B to F is dashed blue. The segment from F to H is dashed blue, and the segment from H to D is dashed blue.
- A callout box shows a table of values and a graph icon. The table is:
- | | | | |
|----|----|----|----|
| 11 | 12 | 21 | 22 |
| 11 | - | 0 | 16 |
| 12 | 0 | - | 21 |
| 21 | 15 | 22 | - |
| 22 | 24 | 16 | 0 |
- The graph icon shows a node connected to a switch, which is connected to another node.
- A warning icon (triangle with exclamation mark) is shown above the callout box.
- Text in the callout box: "To D via POL 1"
- SL 1: < D >, w=3
 - SL 2: < F, D >, w=1
- Text in the callout box: "Topology & Counters"
- Text in the callout box: "CNC"

Demo of DDM with multiple domains

Redistributed and Summarized SRv6 Locators

Coming in CNC8.0 Dec 2026



Dashboard

Topology

Services & Traffic Engineering

Device Management

Alerts

Administration

Topology

Show: Topology

Device groups: Location

Show layers

Location

Auto-focus

Topology

Devices

Links

Family

9 Router

Reachability

9 Reachable

0 Unreachable

0 Unknown

0 Degraded

Filters

9 results

Import

Export

Severity	Reac...	IP address	Host name	Product type	Product.
✓	✓	172.20.48...	P-BOTTOMLEFT	Cisco NCS 540-12...	Route
✓	✓	172.20.48...	P-BOTTOMMID	Cisco NCS 540-12...	Route
✓	✓	172.20.48...	P-BOTTOMRIGHT	Cisco NCS 540-12...	Route
✓	✓	172.20.48...	P-TOPLEFT	Cisco NCS 540-12...	Route
✓	✓	172.20.48...	P-TOPMID	Cisco NCS 540-12...	Route
✓	✓	172.20.48...	P-TOPRIGHT	Cisco NCS 540-12...	Route
✓	✓	172.20.48...	PE-A	Cisco NCS 540-12...	Route
✓	✓	172.20.48...	PE-B	Cisco NCS 540-12...	Route
✓	✓	172.20.48...	PE-C	Cisco NCS 540-12...	Route

Demo of Automated Capacity Planning

Report of congestion with current traffic and simulated single
link/node failure

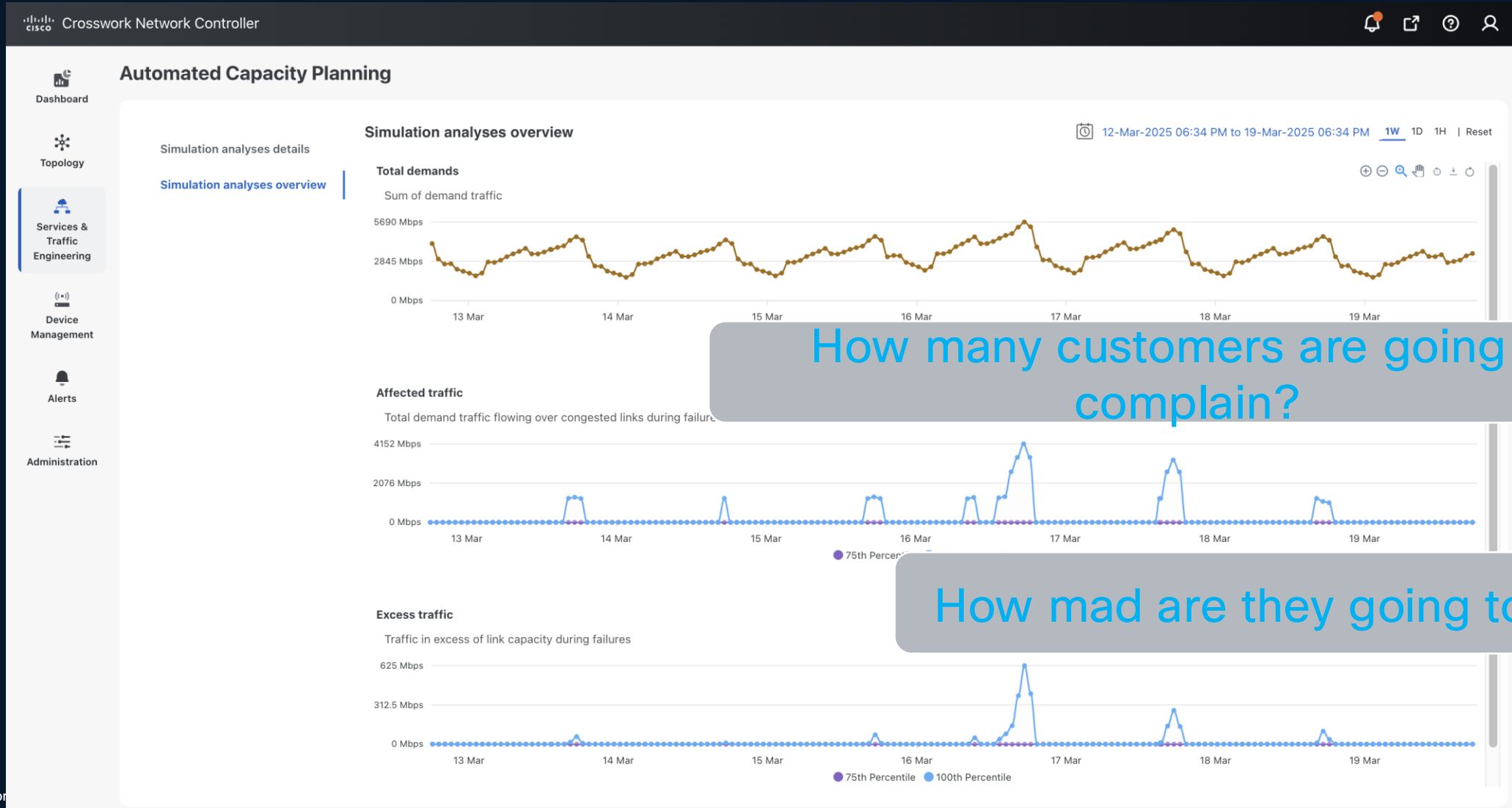
Roadmap – CY2027

Path Query



	Source	Locator prefix	Destination	Algo	Traffic (M...
☐					
☐	NCS540-P...	fccc:cc12:103:...	NCS540-PE-C	0	158.3012
☐	NCS540-P...	fccc:cc12:102:...	PE-B	0	90.7021
☐	NCS540-P...	fccc:cc13:102:...	PE-B	128	167.9225
☐	NCS540-P...	fccc:cc13:103:...	NCS540-PE-C	128	46.6347
☐	NCS540-P...	fccc:cc13:102:...	PE-B	128	121.7083
☐	NCS540-P...	fccc:cc12:102:...	PE-B	0	153.971
☐	NCS540-P...	fccc:cc12:1::/48	NCS540-PE-A	0	89.5224
☐	NCS540-P...	fccc:cc13:1::/48	NCS540-PE-A	128	99.9594
☐	PE-B	fccc:cc12:103:...	NCS540-PE-C	0	144.9968
☐	PE-B	fccc:cc13:1::/48	NCS540-PE-A	128	76.4402
☐	PE-B	fccc:cc13:103:...	NCS540-PE-C	128	138.7437
☐	PE-B	fccc:cc12:1::/48	NCS540-PE-A	0	93.5635

Is there any problem? How bad is it?



How many customers are going to complain?

How mad are they going to be?

What can I do to prevent it? Should I add capacity?

Congested links (occurrences)

Congested links (excess traffic)

Links most likely to be congested due to failures

Link identification

node-22(GigabitEthernet0/0/0/4)	2
node-2(GigabitEthernet0/0/0/5)	1
node-3(GigabitEthernet0/0/0/2)	1
node-5(GigabitEthernet0/0/0/2)	1

Most likely congested links

Congested links (occurrences)

Congested links (excess traffic)

Links that can be the most severely congested due to failures

Link identification

Worst case utilization

(Mbps)

node-2(GigabitEthernet0/0/0/5)	118.35	183.49
node-22(GigabitEthernet0/0/0/4)	109.47	94.66
node-3(GigabitEthernet0/0/0/2)	100.24	2.43
node-5(GigabitEthernet0/0/0/2)	108.3	82.99

Most badly congested links

Or should I add redundancy?

Congested links (occurrences)

Congested links (excess traffic)

Failure scenarios (congestion occurrences)

Failure scenarios (affected traffic)

Failure scenarios most likely to cause saturation

Total 58

Failed object	Type	Count	Number of congested links caused by failed object
node-6		3	
node-21(GigabitEthernet0/0/0/3)		1	
node-23(GigabitEthernet0/0/0/4)		1	
node-13(GigabitEthernet0/0/0/5)			

Failures most likely to cause a congestion

node-14(GigabitEthernet0/0/0/0) - node-1(GigabitEthernet0/0/0/0)	Link
node-14(GigabitEthernet0/0/0/12) - node-1(GigabitEthernet0/0/0/12)	Link
node-14(GigabitEthernet0/0/0/2) - node-12(GigabitEthernet0/0/0/4)	Link
node-14(GigabitEthernet0/0/0/5) - node-15(GigabitEthernet0/0/0/4)	Link

Congested links (occurrences)

Congested links (excess traffic)

Failure scenarios (congestion occurrences)

Failure scenarios (affected traffic)

Failure scenarios that cause most severe congestion

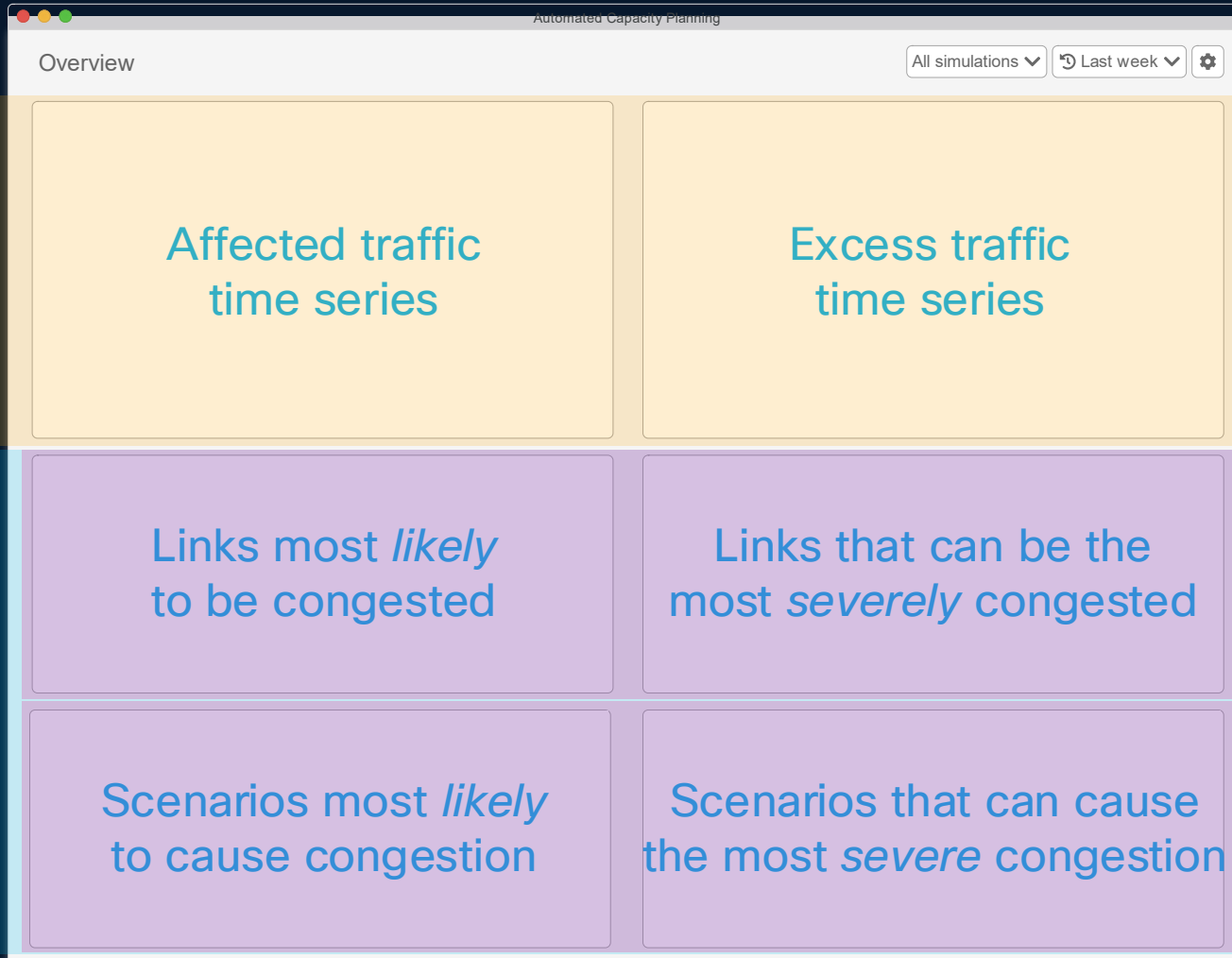
Total 58

Failed object	Type	Congested links (excess traffic)	Affected traffic (%)
node-6			27
node-21(GigabitEthernet0/0/0/3) - node-23(GigabitEthernet0/0/0/4)			29
node-23(GigabitEthernet0/0/0/4) - node-24(GigabitEthernet0/0/0/3)	Link	1225.59	23.89
none	-	0.00	0.00
node-8	Node	0.00	0.00
node-7	Node	0.00	0.00
node-5	Node	0.00	0.00

Failures causing the worse congestions

Making Sense of the Data – 6 ACP KPIs

Is there a problem?
How bad is it?



Add capacity

Add redundancy

Key Takeaways

High-Level Next Steps for SRv6 Automation

2H CY26

CNC 8.0 (Committed)

- Enhanced DDM :
 - SRv6 Locator prefix redistribution and summarization
 - Improved scale and accuracy
- SRv6 Policy counters collection and visualization

1H CY27

CNC 8.1 (Planning)

- SRv6: Real-time hotspot detection (ACP)
- SRv6 Bandwidth Observability Dashboard
- SRv6-aware Local Congestion Mitigation (LCM)
 - Including Automated LCM , Multi-segment list
- TE Agent - Phase 1
 - Any metric, Affinity (incl./excl.), Reserved BW
- Bandwidth Manager
- DDM export to Crosswork Planning

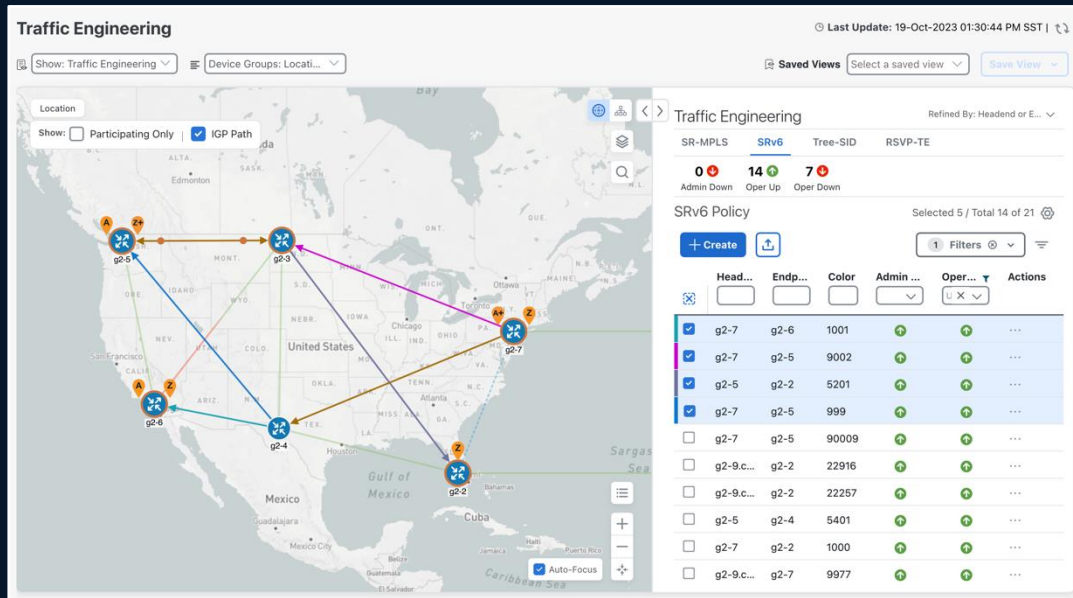
Radar

- Global Congestion Management for SRv6 (Tactical Traffic Engineering)
- TE Agent improvements
 - Disjoint paths, exclude IP, VIA...
- SRv6 Circuit-Style
- Predictive Analytics
- Agentic-AI led recommendations

Some of the features described herein remain in varying stages of development and will be offered on a when-and-if-available basis. This roadmap is subject to change at the sole discretion of Cisco, and Cisco will have no liability for delay in the delivery or failure to deliver any of the products or features set forth in this presentation.

Summary: SRv6 Innovations driving Operational Agility

CNC connects service provisioning, visibility, planning, and congestion mitigation



Challenges

- Realtime, accurate and scalable traffic accounting
- Visibility into network traffic patterns and usage
- Reliable mechanism for traffic engineering and bandwidth optimization

Solution: SRv6 Innovations combined with AI-enabled Crosswork Network Automation

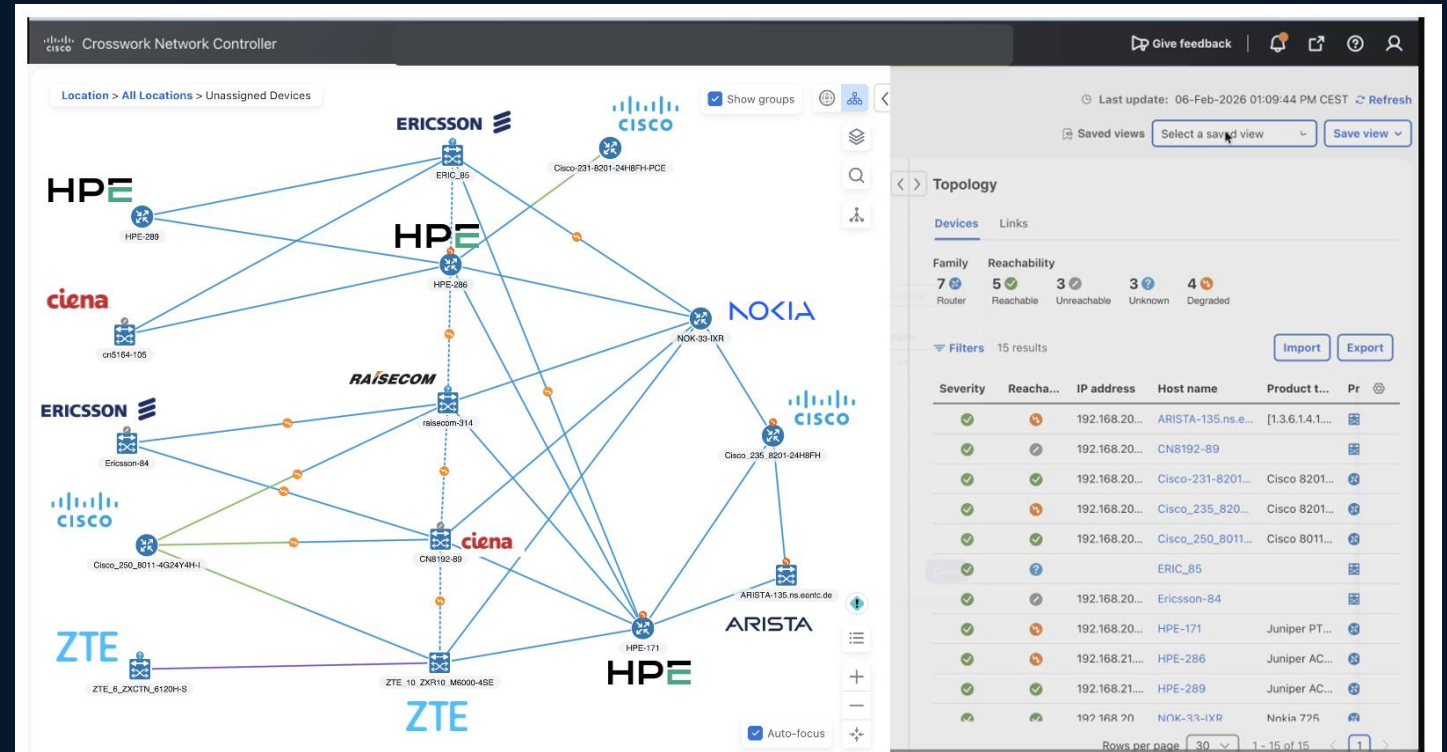
Benefits: Operational Simplicity

- SRv6 demand-matrix counters enables real-time tracking of changing traffic patterns
- Real-time identification of hotspots based on continuous simulation of what-if scenario
- Accurate impact analysis of link or node failures for proactive congestion handling or coordinated maintenance
- Tactical TE for congestion mitigation

References

Crosswork Automation and Assurance – EANTC Interop Highlights – 2026

- Major vendors participated with both Controller and Network Elements:
 - Controller: Cisco, Nokia, ZTE, Keysight.
 - Network Equipment: Cisco, Nokia, ZTE, HPE, Arista, Keysight, Ribbon, Ciena, Raisecom.
- Test cases focused on:
 - PCEP protocol interop
 - BGP-LS protocol interop
 - Netconf/YANG Provisioning
 - Telemetry/gNMI Collection
 - System Inventory
 - Digital Twin
 - EVPN SRv6 Solution
 - 5G xHaul with Segment Routing



- [XR Blog:](https://xrdocs.io/automation/blogs/eantc-2026-cisco-crosswork-automation-demonstrates-multi-vendor-interoperability)
<https://xrdocs.io/automation/blogs/eantc-2026-cisco-crosswork-automation-demonstrates-multi-vendor-interoperability>

- Scan Me



